

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 3 – Comparative Analysis
Weightage:	30% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	CSE AI	Date:	26-08-2026

Code of Conduct

I / L Sai Pavan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: L Sai Pavan

Instructions

- Read each scenario carefully before answering.
- Identify the NLP techniques being compared in the question.
- Analyze each approach based on its working principles, advantages, limitations, and applicability.
- Compare the alternatives using technical reasoning rather than listing definitions.
- Support your analysis with examples from the given scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze sentence representations, compare structural representations of language, and evaluate how different parsing approaches capture meaning and relationships among words.	Q1
Syntax-Driven Semantic Analysis	Analyze parsing strategies for incomplete and ambiguous inputs, evaluate parsing efficiency, and determine suitable methods for real-time language understanding.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze ambiguity in natural language sentences, evaluate ambiguity resolution techniques, and determine the most effective approach for selecting intended meanings.	Q3
Representing Linguistically Relevant Concepts	Analyze grammatical constraints, agreement features, argument structures, and linguistic representations required for accurate language processing.	Q4
Information Retrieval & Syntax-Driven Semantic Analysis	Analyze large-scale parsing strategies, evaluate performance trade-offs, and justify efficient approaches for processing massive linguistic datasets.	Q5

Annexure A – Comparative Analysis

1. A conversational AI system must identify grammatical relationships in user queries containing long-distance dependencies. The developers are comparing constituency parsing and dependency parsing.

Analyze how the two representations differ in capturing hierarchical and word-to-word relationships, and justify which approach is more suitable for extracting meaningful relationships from conversational text.

2. An NLP system processes partially typed search queries such as “best hotels near...” and “book a flight from Chennai to...”. The system currently uses conventional top-down parsing, while the developers are considering Earley parsing.

Analyze how the two approaches behave when input is incomplete or ambiguous, and justify which parsing strategy is more appropriate for interactive NLP applications.

3. A grammar-checking system must distinguish between multiple interpretations of structurally ambiguous sentences. The developers are comparing CFG, PCFG, and neural constituency parsers.

Analyze how each approach represents and resolves syntactic ambiguity, and justify which approach would provide the best balance between interpretability and practical accuracy.

4. A multilingual NLP system must process sentences in which grammatical information depends on number, gender, tense, and person. The developers are considering feature structures and traditional context-free grammar rules.

Analyze how feature structures extend basic grammar representations and justify their usefulness for enforcing grammatical constraints in multilingual systems.

5. A voice assistant must interpret commands containing verbs with different argument requirements, such as “give the book to John,” “sleep,” and “put the file on the desk.” The developers are considering subcategorization frames.

Analyze how subcategorization information helps determine valid sentence structures and justify its importance in semantic and syntactic analysis.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Scope of Items	Comparison Depth	Use of Evidence	Critical Thinking	Clarity of Presentation	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

ANNEXURE A – COMPARATIVE ANALYSIS

Q1. Constituency Parsing vs. Dependency Parsing

Introduction

A conversational AI system must understand grammatical relationships between words, especially when a sentence contains long-distance dependencies. Two important methods for representing sentence structure are **constituency parsing** and **dependency parsing**.

Constituency Parsing

Constituency parsing divides a sentence into hierarchical groups called constituents or phrases. These groups include noun phrases, verb phrases, prepositional phrases, and other grammatical units.

For example, in the sentence:

“The student who joined the course yesterday submitted the assignment.”

Constituency parsing identifies groups such as:

- **The student who joined the course yesterday** – Noun Phrase
- **submitted the assignment** – Verb Phrase
- **the assignment** – Noun Phrase

This representation is useful for understanding the hierarchical organization of a sentence.

Dependency Parsing

Dependency parsing focuses on direct relationships between individual words. The main verb generally acts as the central element, and other words are connected to it through grammatical dependencies.

In the same sentence:

- **student** is related to **submitted** as the subject.
- **assignment** is related to **submitted** as the object.
- **joined** is related to **student** through the relative clause.
- **yesterday** provides a time relationship for **joined**.

Comparison

Aspect	Constituency Parsing	Dependency Parsing
Main focus	Phrase structure	Word-to-word relationships
Representation	Hierarchical	Relational
Basic unit	Phrase/constituent	Individual word
Useful for	Sentence structure and phrase boundaries	Semantic relationships and information extraction

Long-distance relationships	Can represent them through nested structures	Can directly connect related words even when separated
Conversational NLP	Useful for detailed grammatical analysis	Highly useful for extracting relationships and meaning

Analysis

For conversational text, users frequently produce long or complex sentences. Important words may be separated by several other words. Dependency parsing can directly represent relationships between words regardless of their physical distance in the sentence.

For example, in a sentence where the subject and its verb are separated by a relative clause, dependency parsing can still establish the subject–verb relationship clearly.

Conclusion

Both approaches are valuable. Constituency parsing is better for understanding **hierarchical phrase structure**, while dependency parsing is generally more suitable for **extracting meaningful word-to-word relationships** from conversational text.

Therefore, for a conversational AI system focused on semantic relationship extraction, **dependency parsing is the more suitable primary representation**, although constituency information can be useful as an additional source of syntactic information.

Q2. Top-Down Parsing vs. Earley Parsing for Incomplete Queries

Introduction

Interactive NLP systems must process partially typed and incomplete queries. Examples include:

- “**best hotels near...**”
- “**book a flight from Chennai to...**”

The parser must be able to work with incomplete information rather than waiting for a perfectly formed sentence.

Conventional Top-Down Parsing

Top-down parsing starts with a sentence-level grammar rule and attempts to construct the sentence by predicting the required components.

For example, after receiving:

“**Book a flight from Chennai to...**”

the parser may expect a destination after the word “**to.**”

If the input ends at this point, the parser has an incomplete structure.

Top-down parsing may also encounter excessive backtracking when several grammar rules are possible.

Earley Parsing

Earley Parsing is designed to handle general context-free grammars and can process input incrementally.

For:

“best hotels near...”

the parser can maintain the incomplete structure and recognize that a location is expected after **“near.”**

For:

“book a flight from Chennai to...”

it can recognize that the destination is missing without treating the entire query as invalid.

Comparison

Aspect	Top-Down Parsing	Earley Parsing
Complete sentences	Suitable	Suitable
Incomplete input	Less effective	Highly suitable
Ambiguity	May require backtracking	Maintains multiple possibilities
Interactive systems	Can become inefficient	Well suited
Partial input	Limited support	Strong support
Complex structures	May require repeated processing	Handles general CFG structures effectively
Real-time applications	Can experience delays	Better suited to incremental processing

Analysis

In interactive search systems, users do not always complete their queries in one step. They may pause, edit, or continue typing.

Earley Parsing is more appropriate because it can preserve partial parsing states and continue processing as additional words arrive.

For example:

“book a flight from Chennai to...”

does not need to be discarded. The system can identify that the destination is still expected and wait for further input.

Conclusion

Earley Parsing is more appropriate for interactive NLP applications because it handles **incomplete input**,

Q3. CFG vs. PCFG vs. Neural Constituency Parsing

Introduction

Natural language often contains structural ambiguity, meaning that the same sentence can have more than one syntactic interpretation.

A grammar-checking system can use CFG, PCFG, or neural constituency parsing to analyse such sentences.

Context-Free Grammar

A CFG represents valid grammatical structures using grammar rules.

Its major advantage is **interpretability** because the rules are explicitly defined.

However, when multiple parse trees are possible, a basic CFG does not inherently determine which interpretation is the most likely.

Probabilistic Context-Free Grammar

PCFG extends CFG by assigning probabilities to grammar rules.

If a sentence has several possible structures, the system can compare their probabilities and select the most likely parse.

For example, if one interpretation is frequently observed in training data, it can receive a higher probability.

Thus:

CFG → Generates possible structures

PCFG → Generates and ranks possible structures

Neural Constituency Parser

Neural constituency parsers use machine-learning models to learn syntactic patterns from large amounts of data.

They can capture complex linguistic patterns and contextual information that may be difficult to encode manually.

Their main advantages include:

- High practical accuracy.
- Strong contextual understanding.
- Ability to learn complex patterns.
- Better handling of naturally occurring language.

However, neural systems can be less interpretable than rule-based approaches.

Comparison

Aspect	CFG	PCFG	Neural Constituency Parser
Grammar representation	Explicit rules	Rules with probabilities	Learned representation
Ambiguity handling	Generates alternatives	Ranks alternatives	Uses learned contextual information
Interpretability	Very high	High	Lower
Context sensitivity	Limited	Moderate	High
Practical accuracy	Moderate	Better than basic CFG	Generally strongest
Data requirement	Low	Training data useful	Usually requires substantial training data
Explainability	Excellent	Good	More difficult

Analysis

For a grammar-checking system, interpretability is important because users may want to know why a particular structure was selected.

A basic CFG provides strong interpretability but limited ambiguity resolution.

A PCFG provides a good compromise because it maintains explicit grammar rules while using probabilities to select the most likely interpretation.

Neural constituency parsing can provide excellent practical accuracy but may be harder to explain.

Conclusion

If the objective is the **best practical accuracy**, a neural constituency parser is generally the strongest choice.

However, for a system requiring a **balance between interpretability and practical accuracy**, **PCFG is a strong choice**. A modern system can also combine neural models with rule-based or probabilistic grammatical constraints.

Q4. Feature Structures vs. Traditional CFG Rules

Introduction

Multilingual NLP systems must handle grammatical information such as:

- Number.

- Gender.
- Person.
- Tense.
- Case and other language-specific grammatical properties.

A basic CFG does not automatically encode all of these constraints.

Traditional CFG

A traditional CFG describes grammatical structures using production rules.

For example, it can specify that a sentence contains a noun phrase and a verb phrase.

However, basic CFG rules may allow structures that have incorrect agreement.

For example:

“The students is studying.”

The subject is plural, while the verb is singular.

A simple CFG may require additional grammar rules to prevent such mismatches.

Feature Structures

Feature Structures attach attributes and values to linguistic units.

For example, a noun may contain features such as:

- Number = plural
- Person = third
- Gender = feminine

A verb can contain corresponding features.

The parser can then check whether the features are compatible.

For example:

Students → Number: Plural

Study → Number: Plural

The features agree.

Comparison

Aspect	Traditional CFG	Feature Structures
Basic grammatical structure	Yes	Yes
Number agreement	Limited unless explicitly encoded	Directly represented
Gender agreement	Difficult with simple rules	Easily represented
Person agreement	Requires additional rules	Explicitly represented

Tense information	Limited	Can be included as a feature
Multilingual processing	Less flexible	More flexible
Grammar constraints	Rule-based	Rule + feature-based

Importance in Multilingual Systems

Different languages have different grammatical requirements.

For example, some languages have extensive gender and case agreement, while others have different agreement patterns.

Feature Structures allow these properties to be represented explicitly without creating a very large number of separate grammar rules.

This makes grammar representation more flexible and maintainable.

Conclusion

Feature Structures extend CFG by adding linguistic attributes and constraints. They are particularly useful for multilingual NLP because they help enforce **number, gender, person, tense, and other grammatical relationships**.

Therefore, Feature Structures provide a more expressive representation than basic CFG rules for multilingual language processing.

Q5. Subcategorization Frames in Voice Assistants

Introduction

Different verbs require different numbers and types of arguments. This property is important when interpreting voice commands.

Consider the following examples:

1. **“Give the book to John.”**
2. **“Sleep.”**
3. **“Put the file on the desk.”**

These verbs have different argument requirements.

“Give”

The verb **give** normally requires:

- A giver.
- An object being given.
- A recipient.

In the sentence:

“Give the book to John.”

The semantic roles are:

- Giver = understood speaker or user.
- Object = book.
- Recipient = John.

Therefore, the verb requires multiple arguments.

“Sleep”

The verb **sleep** can be used without an object.

In:

“Sleep.”

the command is complete even though there is no direct object.

Therefore, **sleep** has a different argument requirement from **give**.

“Put”

The verb **put** normally requires:

- An object being placed.
- A location.

For example:

“Put the file on the desk.”

The roles are:

- Object = file.
- Location = desk.

The phrase **“on the desk”** is therefore an important argument of the verb.

Comparison

Verb	Required Information	Example
Give	Giver, object, recipient	Give the book to John
Sleep	Person performing the action	Sleep
Put	Object and location	Put the file on the desk

Importance of Subcategorization Frames

A subcategorization frame describes the types of arguments that a verb can or must take.

This helps the NLP system determine whether a sentence is structurally and semantically appropriate.

For example, if the user says:

“Put the book.”

the system can recognize that the command may be incomplete because the verb **put** normally requires a location.

It can then ask:

“Where would you like me to put the book?”

Similarly, if the user says:

“Give the book.”

the system may recognize that a recipient is missing and request additional information.

Role in Semantic Analysis

Subcategorization frames connect syntax with meaning.

They help determine:

- What arguments a verb requires.
- Which phrases represent objects.
- Which phrases represent recipients.
- Which phrases represent locations.
- Whether an input is complete.
- Whether the user's command can be executed.

Conclusion

Subcategorization information is essential for voice assistants because it allows the system to understand the argument requirements of different verbs. It improves both syntactic parsing and semantic interpretation and helps the assistant identify missing information before carrying out an action.

Overall Conclusion

The five comparative analyses demonstrate that no single parsing technique is sufficient for every NLP application.

Constituency parsing provides a strong representation of **hierarchical sentence structure**, while dependency parsing is particularly effective for **word-to-word relationships**.

For incomplete and interactive queries, **Earley Parsing** is more appropriate than conventional top-down parsing because it can maintain partial and ambiguous parsing states.

For ambiguity resolution, **CFG** provides interpretability, **PCFG** adds probabilistic ranking, and **neural constituency parsers** provide strong practical accuracy. A combination of these approaches can be particularly effective.

Feature Structures extend grammatical representations by explicitly encoding number, gender, person, and tense, making them highly valuable for multilingual NLP.

Finally, **subcategorization frames** help voice assistants understand the argument requirements of verbs and identify incomplete or incorrect commands.

Overall, the most effective NLP systems combine syntactic structure, probabilistic information, linguistic features, and semantic knowledge rather than relying on a single parsing technique.